220 ["In personality they . . . I chose Breit."] Wheeler (1979). This reference is an autobiographical account of Wheeler's research in nuclear physics.

220, 222 [Wheeler and Bohr at Princeton . . . The Bohr–Wheeler article . . . *Physical Review*] Bohr and Wheeler (1939), Wheeler (1979). Bohr and Wheeler did not name plutonium-239 by name in their paper, but Louis A. Turner inferred directly from their Figure 4 that it was an ideal nucleus for sustaining chain reactions, and proposed in a famous classified memorandum that it be used as the fuel for the atomic bomb (Wheeler, 1985).

223 [Zel'dovich and a close friend, . . . for all the world to see.] INT-Zel'dovich, Zel'dovich and Khariton (1939).

223 [Wheeler was the lead scientist . . . Nagasaki bomb.] For some details of Wheeler's key role, see pages 2–5 of Klauder (1972).

223 ["If atomic bombs . . . at Los Alamos and Hiroshima."] From a speech by Oppenheimer at Los Alamos, New Mexico, on 16 October 1945; see page 172of Goodchild (1980).

223 ["In some sort of crude sense . . . cannot lose."] Page 174 of Goodchild (1980).

223–224 ["As I look back . . . August 6, 1943."] Wheeler (1979).

225 [While this massive effort. . . over to the American design.] These details were revealed by Khariton in a lecture in Moscow, which was reported in the *New York Times* of Thursday, 14 January 1993, page A5.

225 [accumulation of waste . . . square miles of countryside.] Medvedev (1979).

226–227 ["We base our recommendations . . . genocide."] Report of 30 October 1949 from the General Advisory Committee to the U.S. Atomic Energy Commission. Reproduced in the appendix of York (1976).

227 ["Nine out of ten . . . stroke of genius."] Bethe (1982).

227 [As Wheeler recalls, "We did an immense amount . . . get things out."] INT-Wheeler.

228 [Wheeler recalls, "While I was starting . . . on the project."] INT-Wheeler.

229 ["The program we had in 1949 . . . once you had it."] USAEC (1984), p. 251.

229 ["I'm told . . . thermonuclear devices"] INT-Wheeler.

229 [In spring 1948, fifteen months before] There seems to be some confusion over the date on which the Soviet H-bomb design work was initiated. Sakharov (1990) dates it as spring 1948, but Ginzburg (1990) dates it as 1947.

229 [In June 1948, a second superbomb team] This is the date given by Sakharov (1990); Ginzburg (1990) places the date in 1947.

229 Footnote 3: Sakharov's speculation is outlined in Sakharov (1990). Zel'dovich's assertion was made verbally to close Russian friends, who transmitted it to me.

230 ["Our job is to lick Zel'dovich's anus."] Quoted to me by Vitaly Ginzburg, who was present. Sakharov was also present; in the English version of his memoirs (Sakharov, 1990), the quotation is expressed as "Our job is to kiss Zel'dovich's ass." For some of my own views on the complex relationship between Zel'dovich and Sakharov, see Thorne (1991).

230 ["that bitch, Zel'dovich."] This quote by Landau has been passed on to me independently by several Soviet theoretical physicists.

230 [Sakharov proposed . . . lithium deuteride (LiD).] Romanov (1990).

231 [it was 800 times more powerful. . . Hiroshima.] The numbers I cite for the energy release in various bomb explosions are taken from York (1976).

231 ["I am under the influence . . . his humanity."] Sakharov (1990).

232 [In March 1954, Sakharov . . . Teller–Ulam idea,] Romanov (1990), Sak harov (1990). Romanov, in an article in honor of Sakharov, attributes the discovery jointly to Sakharov and Zel'dovich. Sakharov says that "[s]everal of us in the theoretical departments came